Exercises week 4

8CM10 - Introduction to Quantum Physics and Structure of Matter

Atoms

1. Electrons in atoms are known to have kinetic energies in the range of a few eV. Show that the uncertainty principle allows electrons of this energy to be confined in a region the size of an atom (0.1 nm)

Bohr Model

2. In the $n = 3$ state of hydrogen, find the electron's velocity, kinetic energy, and potential energy.

3. An electron is in the $n = 5$ state of hydrogen. To what states can the electron make transitions, and what are the energies of the emitted radiations?

4. Use the Bohr formula to find the energy differences $E(n_1 \rightarrow n_2) = E_{n_1} - E_{n_2}$ and show that:
   a. $E(4 \rightarrow 2) = E(4 \rightarrow 3) + E(3 \rightarrow 2)$
   b. $E(4 \rightarrow 1) = E(4 \rightarrow 2) + E(2 \rightarrow 1)$

5. Find the shortest and the longest wavelengths of the Lyman series of singly ionized helium.

6. Draw an energy-level diagram showing the lowest four levels of singly ionized helium. Show all possible transitions from the levels and label each transition with its wavelength.

The Franck-Hertz Experiment

7. A hypothetical atom has only two excited states, at 4.0 and 7.0 eV, and has a ground-state ionization energy of 9.0 eV. If we used a vapor of such atoms for the Franck-Hertz experiment, for what voltages would we expect to see decreases in the current? List all voltages up to 20 V.